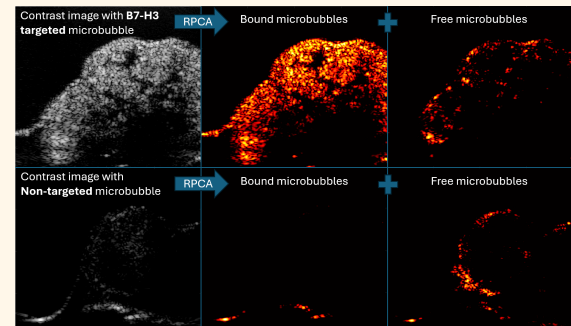


Enhancing Ultrasound Molecular Imaging: Toward Real-Time RPCA-Based Filtering to Differentiate Bound and Free Microbubbles

Hoda S. Hashemi, Dongwoon Hyun, Nathan Nguyen, Jihye Baek, Arutselvan Natarajan, Farbod Tabesh, Andrew Andrzejek, Ramasamy Paulmurugan, and Jeremy J. Dahl

Abstract—Ultrasound molecular imaging (UMI) is an advanced imaging modality that shows promise in detecting cancer at early stages. It uses microbubbles as contrast agents, which are functionalized to bind to cancer biomarkers overexpressed on endothelial cells. A major challenge in UMI is isolating bound microbubble signal, which represents the molecular imaging signal, from that of free-floating microbubbles, which is considered background noise. In this work, we propose a fast GPU-based method using robust principal component analysis (RPCA) to distinguish bound microbubbles from free-floating ones. We explore the method using simulations and measure the accuracy using the Dice coefficient and RMS error as functions of the number of frames used in RPCA reconstruction. Experiments using stationary and flowing microbubbles in tissue-mimicking phantoms were used to validate the method. Additionally, the method was applied to data from ten transgenic mouse models of breast cancer development, injected with B7-H3 targeted microbubbles, and two mice injected with non-targeted microbubbles. The results showed that RPCA using 20 frames achieved a Dice score of 0.95 and a computation time of 0.2 seconds, indicating that 20 frames is potentially suitable for real-time implementation. On in vivo data, RPCA using 20 frames achieved a Dice score of 0.82 when compared with a conventional preclinical method, Differential Targeted Enhancement (DTE), indicating good agreement between the two, given the limitations of each method.

Index Terms—Ultrasound molecular imaging, Microbubbles, Contrast agents, Robust principle component analysis, RPCA



I. INTRODUCTION

ULTRASOUND molecular imaging (UMI) is an advanced imaging modality that uses microscopic gas-filled bubbles as contrast agents to improve the detection and visualization of biological structures and processes. These microbubbles, typically 1–5 μm in size, efficiently enhance ultrasound wave reflections, enabling high-contrast imaging. Many disease processes, including precursor lesions and invasive cancer [1], [2], can be defined by the expression of specific molecules on the endothelial surface. These molecules can be targeted using special ligands attached to the microbubble shells. Through the ligand, the microbubble is bound to the diseased tissue,

enabling imaging of the non-physical or molecular properties of the tissue.

A significant challenge in UMI lies in distinguishing the signal from tissue-bound targeted microbubbles from background interference, including tissue-leakage signals, noise, and free-floating microbubbles. Contrast enhanced ultrasound (CEUS), such as pulse inversion and amplitude modulation, [3]–[5], cannot distinguish tissue leakage signals from bubble signals, and it does not separate the molecular signals of bound bubbles from background noise, including free-floating bubbles. Differential targeted enhancement (DTE) was proposed to overcome the challenges in CEUS imaging [6], [7]. It involves using a high-intensity ultrasound pulse to burst the microbubbles, which enables a comparison between pre- and post-burst CEUS images [8]. However, this method has notable limitations: it destroys the contrast agents, thereby leaving “holes” in subsequent images; cannot be performed in real-time because it requires collecting and averaging multiple ultrasound images before and after the destruction (burst) of microbubbles; and it necessitates precise tumor localization, as it captures images from a single field of view, which restricts

This work was supported by the Stanford Cancer Imaging Training (SCIT) program, funded by the National Cancer Institute (NCI) through training grant T32-CA009695, and the National Institute of Biomedical Imaging and Bioengineering (NIBIB) under grant R01-EB031799.

H. S. Hashemi, N. Nguyen, J. Baek, A. Natarajan, F. Tabesh, A. Andrzejek, R. Paulmurugan, and J. J. Dahl are with the Department of Radiology, Stanford University, Stanford, CA 94305 USA.

D. Hyun is with the Siemens Healthineers, Palo Alto, CA 94304 USA.

H. S. Hashemi (e-mail: hodah@stanford.edu) and J. J. Dahl (e-mail:jjdahl@stanford.edu) are the corresponding authors.

Highlights

- **Ultrasound molecular imaging has shown promising results in early cancer detection using targeted microbubbles that can bind to cancer biomarkers.**
- **This paper presents a novel, nondestructive method based on robust principal component analysis to distinguish disease-indicating bound microbubbles from free microbubbles circulating with blood flow.**
- **The proposed method was tested on simulation, phantom, and in vivo data from mouse models of breast cancer, successfully separating bound and free microbubbles in ultrasound time-series data.**

the broader applicability of UMI [9].

Previous studies employed various techniques to overcome the limitations of DTE imaging, including dwell-time imaging [10] and minimum intensity projection [11], to detect bound microbubbles through temporal analysis. Although these methods enhance the specificity of bound microbubble detection, dwell-time imaging is limited by its sensitivity to the system noise level and when multiple microbubbles occupy the same voxel, while the minimum intensity projection approach depends on offline processing and requires accurate motion compensation and tracking to avoid false or missed detections. Additionally, singular value decomposition (SVD) based methods have been utilized [5], [12], [13]. SVD and principal component analysis (PCA) are closely related to least squares procedures in their mathematical formulation, making them highly sensitive to outliers. In extreme cases, outliers can distort the computed principal components and singular vectors. The Robust Principal Component Analysis (RPCA) models data as the sum of a low-rank matrix, representing the underlying clean data, and a sparse matrix, capturing outliers and noise, making it more robust to outliers compared to SVD and PCA.

RPCA has been widely used in computer science for identifying robust principal components. Depending on the specific application, the emphasis may be placed on either the low-rank component or the sparse component. RPCA has been utilized in ultrasound localization microscopy (ULM) to enhance clutter filtering and noise suppression. The ULM approach focuses on extracting moving microbubbles and reconstructing vessels by leveraging the sparse matrix component of RPCA [14]. We recently introduced the application of the RPCA algorithm in UMI to localize stationary targeted microbubbles bound to cancer biomarkers, while distinguishing free-floating microbubbles and provided some preliminary results [15].

In this work, we extend our work in [15] to implement a fast GPU-based version and improve our analysis by validating RPCA filtering on simulated and experimental-phantom ultrasound data. The method is demonstrated on in vivo data from transgenic mouse models of breast cancer where B7-H3 targeted microbubbles are injected via the tail vein. The results of the proposed method are compared with the DTE imaging, and its potential for real-time implementation is analyzed.

II. METHODS

A. Overview of Proposed Robust Principle Component Analysis (RPCA) Algorithm

For an ultrasound time-series consisting of N_t image frames collected sequentially over time, a two-dimensional spatiotem-

poral matrix X is constructed by reshaping each frame into a vector and arranging these vectors as the columns of X . Given that each ultrasound frame has dimensions $m \times n$, the resulting matrix has a size of $mn \times N_t$. The echo data in X is modeled as a combination of signals originating from bound microbubbles, free microbubbles, and noise. Bound microbubbles, along with possible tissue leakage, exhibits significant spatiotemporal coherence, and are modeled via a low-rank matrix L , as high spatiotemporal coherence often implies redundancy and linear dependence among columns, resulting in a low-rank representation. Conversely, the movement of free microbubbles, along with thermal noise, exhibit low spatiotemporal coherence and are modeled via a sparse matrix S , where the columns are sparse due to the transient and unstructured nature of these components. Consequently, the echo data can be expressed as

$$X = L + S. \quad (1)$$

It has been shown theoretically [16] that Principal Component Pursuit (PCP), a convex optimization technique, can effectively decompose X into its low-rank and sparse components by solving the optimization problem

$$\begin{aligned} &\text{minimize } \|L\|_* + \lambda \|S\|_1 \\ &\text{subject to } L + S = X \end{aligned} \quad (2)$$

where $\|L\|_* = \sum_i \sigma_i(L)$ is the nuclear norm of L , representing the sum of its singular values, and $\|S\|_1 = \sum_{ij} |S_{ij}|$ is the l_1 -norm of S , representing the sum of the absolute values of its elements. The parameter λ serves as a regularization term, balancing the trade-off between the low-rank approximation and the sparse component in the matrix decomposition. It has been demonstrated that the choice $\lambda = \frac{1}{\sqrt{\max(n_1, n_2)}}$ universally and consistently yields the correct solution, where n_1 and n_2 are the dimensions of the matrix X . While this may not be immediately apparent, mathematical analysis confirms the validity of this selection [16].

A widely used approach to solve PCP is the Alternating Direction Method of Multipliers (ADMM). To optimize equation (2), ADMM forms a new objective function, known as the augmented Lagrangian function ($\mathcal{L}$), given by:

$$\begin{aligned} \mathcal{L}(L, S, Y) = &\|L\|_* + \lambda \|S\|_1 + \langle Y, X - L - S \rangle \\ &+ \frac{\mu}{2} \|X - L - S\|_F^2 \end{aligned} \quad (3)$$

This objective function includes the original objective function from equation (2), along with a linear term $\langle Y, X - L - S \rangle$ representing the inner product between the Lagrange multiplier

Y and the constraint, and a quadratic penalty term $\|X - L - S\|_F^2$, weighted by the parameter μ . Both terms help enforce the constraint as the iterations progress [16]. ADMM solves equation (3) iteratively by repeating three steps:

- 1) Minimize $\mathcal{L}$ with respect to L (fixing S and Y): It has been shown [16] that minimizing the augmented Lagrangian in Equation (3) with respect to L is equivalent to performing thresholded singular value decomposition (SVD) with a threshold of $\tau = \frac{1}{\mu}$, where the largest singular values are retained in L while the smaller ones are set to zero, allowing L to capture the dominant structures in the data. This can be defined as:

$$\min_L \mathcal{L}(L, S, Y) = D_{\frac{1}{\mu}}(X - S + \mu^{-1}Y) \quad (4)$$

where the singular value thresholding operator for any matrix M is given by $D_{\tau}(M) = U\Lambda_{\tau}V^*$, with $\Lambda_{\tau} = \max(\Lambda - \tau, 0)$ applying the threshold $\tau = \frac{1}{\mu}$ to the singular values of M . Here, $M = U\Lambda V^*$ represents the SVD of M , where Λ is the diagonal matrix of singular values.

- 2) Minimize $\mathcal{L}$ with respect to S (fixing L and Y): It has been shown [16] that this minimization problem is equivalent to element-wise thresholding of the residual, $X - L$, meaning that small-magnitude elements are set to zero while large-magnitude elements remain. In this way, S can capture the outliers. This has been defined by the shrinkage operator:

$$S = \text{shrink}_{\frac{\lambda}{\mu}}(X - L + \mu^{-1}Y) \quad (5)$$

where the shrinkage operator is defined as $\text{shrink}_{\gamma}[x] = \text{sgn}(x) \max(|x| - \gamma, 0)$ where x represents an individual element of the matrix $X - L + \mu^{-1}Y$. This means that for each entry x_{ij} in $X - L + \mu^{-1}Y$, it is thresholded according to $\gamma = \frac{\lambda}{\mu}$: if x_{ij} is greater than γ , it is reduced by γ ; otherwise, it is set to zero. The output will be the updated sparse matrix S , which contains the thresholded values.

- 3) Update Lagrangian multiplier Y : In practice, since the updates for L and S in the first two steps are performed separately, inconsistencies may arise such that $L + S$ does not exactly equal X . The Lagrange multiplier Y helps gradually correct these discrepancies. The update for Y is given by:

$$Y_{k+1} = Y_k + \mu(X - L_k - S_k) \quad (6)$$

As suggested in [17], ADMM works well with the choice $\mu = \frac{n_1 n_2}{4\|X\|_1}$ where n_1 and n_2 are the dimensions of the data matrix X . We terminate the algorithm when $\|X - L - S\|_F < \delta\|X\|_F$. We selected the parameter $\delta = 10^{-9}$ empirically by visually inspecting the L image to ensure that it is stationary, meaning that all free bubbles are captured in S . The subscript F in $\|\cdot\|_F$ refers to the Frobenius norm of a matrix where, for matrix X , is defined as: $\|X\|_F = \sqrt{\sum_{i,j} |X_{i,j}|^2}$.

The resulting matrices L and S have the same size as matrix X , i.e., $mn \times N_t$, where N_t is the number of frames over time. The matrices L (bound microbubbles) and S (moving

microbubbles) are then reshaped to size $m \times n \times N_t$, corresponding to the frames of the original time-series ultrasound data.

B. GPU Implementation

We have adapted a GPU-based implementation of the RPCA method using PyTorch to efficiently decompose ultrasound molecular imaging data into low-rank and sparse components [18]. This implementation leverages CUDA acceleration to handle large datasets while maintaining computational efficiency. The input RF data is normalized and reshaped to form the data matrix X before decomposition. The regularization parameter λ is set as explained earlier and μ is initialized as $\mu = \frac{n_1 n_2}{4\|X\|_1}$, where n_1 and n_2 are the dimensions of each RF frame. We initialize and update L , S , and the dual variable Y iteratively until convergence, using the Frobenius norm as the stopping criterion. The dominant operation in each iteration of RPCA is the SVD, with a complexity of $O(mn \times N_t \times \min(mn, N_t))$ per iteration for a matrix X of size $mn \times N_t$. To manage memory usage, the RPCA decomposition is performed in PyTorch using CUDA tensors. The major variables X , L , S , and Y are pre-allocated on the GPU to minimize host-device transfer overhead. Intermediate results and SVD operations are performed in-place where possible to limit peak memory usage, and we use `torch.cuda.empty_cache()` as needed between iterations to ensure efficient memory reuse. For efficient matrix operations, all thresholding and shrinkage operations are vectorized, and SVD is applied only once per iteration using PyTorch's optimized linear algebra backend.

C. Simulation

1) *Simulations for Microbubble Imaging*: Ten simulations were performed using a random number and location of stationary and moving microbubbles in the image plane. For the total bound and free microbubbles, a random number between 800 and 1600 microbubbles were selected and randomly spread over a $6 \times 5 \text{ mm}^2$, resulting in an average density of 27 to 54 microbubbles/ mm^2 . This distribution of microbubbles included both bound and free microbubbles. The simulations were performed in Field II [19], [20], using the Marmottant model [21] to represent the nonlinear frequency response of microbubbles. Realistic microbubble responses to incident ultrasonic pulses were simulated using BubbleSim [22], and the resulting acoustic backscatter was modeled using a simulated L12-3v ultrasound transducer in Field II. The tissue simulations are performed separately from the microbubble simulations, and their RF signals are then combined using superposition. Thermal noise is subsequently added at -12 dB relative to the root-mean-square (RMS) of the B-mode signal. In the case of contrast mode, only the RF signals of stationary and moving microbubbles are combined with the thermal noise. For each simulation, a time series of harmonic images consisting of 60 frames was generated and served as the input for the RPCA algorithm. Angled plane wave transmissions from 25 different angles were simulated modeling a Verasonics L12-3v transducer (Kirkland, WA, USA) and coherently compounded via delay-and-sum beamforming

to produce synthetically focused radiofrequency (RF) data. Table I presents the parameters used in the Field II simulation and BubbleSim software.

Parameter	Value
Transducer	Linear (L12-3v)
Number of elements	128
Pitch [mm]	0.2
Center frequency [MHz]	8
Bandwidth [MHz]	[3, 12]
TX angular range	$[-5^\circ, 5^\circ]$
TX angle spacing	0.4°
Bubble radius [μm]	1
Internal equilibrium pressure [Pa]	1.013×10^5
Density of surrounding liquid [kg/m^3]	1000
Viscosity of the liquid [$\text{Pa} \cdot \text{s}$]	10^{-3}
Speed of sound in the liquid [m/s]	1500
Thermal conductivity of the gas [W/mK]	26.2×10^{-3}
Density of gas [kg/m^3]	1.161
Heat capacity of the gas [J/kgK]	1.007×10^3
Adiabatic constant of the gas	1.4

TABLE I: Simulation parameters.

2) Modeling of microbubble motion: The motion of microbubbles is governed by Newton's second law, with forces such as fluid drag, acoustic radiation force, and buoyancy potentially influencing their trajectories. While these forces can affect bubble motion under high acoustic pressures or over long observation times, the simulations in this study were performed assuming low mechanical index values, where radiation and buoyancy effects are expected to be negligible compared to stochastic flow dynamics. Therefore, in our model, bubble motion was simplified by considering only random accelerations and velocities. Each bubble is assigned an initial velocity amplitude, v_0 which is randomly selected from a uniform distribution in the range of $[0.01, 2]$ mm/s, ensuring realistic microbubble speeds. The velocity components in the axial and lateral directions are determined by multiplying v_0 with the cosine and sine of a randomly chosen angle within $[-\pi, \pi]$, allowing bubbles to start in random directions. To introduce variability in bubble trajectories, a random acceleration within the range of $[1, 10]$ mm/s² is applied at each time step. The acceleration components are obtained by multiplying this value by the cosine and sine of a random angle within $[-\pi, \pi]$, ensuring a reasonable spread in different directions.

At each time step, velocities and new positions are updated using the following equations in the axial and lateral directions separately: $v(t) = v(t-1) + a \cdot dt$, and $x(t) = x(t-1) + v(t) \cdot dt$. Based on this model, the moving bubbles start at random positions and directions and follow curved paths due to random accelerations applied over time. Some bubbles exhibit more deviation, while others have relatively straight trajectories depending on their initial conditions and variations in acceleration magnitude and direction. The presence of a time-dependent velocity update suggests a gradual shift in trajectory rather than sudden changes.

To further investigate the effect of microbubble velocity on the performance of the proposed method, ten simulations were conducted, each with four different velocities for the free microbubbles. In each simulation, all free microbubbles were

assigned the same initial velocity but random directions. The velocities used were 0.02, 0.2, 2, 20 mm/s, respectively. This range of values covers blood velocities in mice and humans, spanning both capillaries and vessels.

3) Singular Value Decomposition (SVD): To provide benchmark values for the simulation results, the SVD algorithm is applied to the simulation data. The algorithm retains the largest k singular values, where k is chosen such that the retained singular values account for 95% of the total energy. The energy ratio is defined as: $\frac{\sum_{i=1}^k \sigma_i^2}{\sum_{i=1}^r \sigma_i^2}$, where σ_i are the singular values, and r is the full rank. The smallest k is selected such that the energy ratio is higher than 95%.

D. Phantom Experiment

An agar phantom (2% agarose, 2% graphite) with a 2 mm flow channel was created to mimic a vessel within tissue. One end of the channel was closed. To simulate stationary bubbles, we created custom non-targeted microbubbles according to the protocol described in [23], diluted the microbubbles with saline and mixed them with ultrasound gel to enhance their stability, and then injected them into the open end of the channel. After injection, the open side of the channel was also closed, and the microbubbles were allowed to settle and become stationary. Time-series beamformed RF data consisting of 20 frames were captured using an ACUSON S3000 ultrasound scanner (Siemens Medical Solutions, Mountain View, CA, USA) in the Contrast Mode at a frame rate of 20 frames/s with a MI of 0.07. The conventional focused transmit data were acquired using a 9L4 linear transducer at 5 MHz center frequency and in the Research Mode [24]. To simulate moving microbubbles, they were first diluted in saline to a final concentration of 10^7 microbubbles/mL. A peristaltic pump (ISMATEC, IDEX Corporation, IL, USA) and its tubing were connected to both ends of the phantom channel to create a closed-loop system for continuous microbubble circulation. A flow rate of 3.77 mL/min was used, corresponding to a velocity of approximately 20 mm/s in tubing with a 2 mm diameter, to approximate flow conditions in small tumor-feeding vessels. Reported microvascular (capillary/precapillary) velocities in human tissue are typically much lower (sub-mm/s to a few mm/s) [25], [26], while Doppler measurements of tumor feeding vessels commonly report peak velocities on the order of tens to a few hundred mm/s; tumor flow is known to be spatially heterogeneous. Thus 20 mm/s represents an intermediate, physiologically relevant value appropriate for phantom validation [27].

E. In Vivo Data Collection

All animal experiments were approved by the Institutional Administrative Panel on Laboratory Animal Care at Stanford University under protocol 14686 and adhered to the ARRIVAL 2.0 guidelines. Ten transgenic mouse models of breast cancer development [28] were used to collect in vivo UMI data. Imaging was performed using B7-H3 targeted microbubbles. B7-H3 is a glyco-protein belonging to the B7 family of immune checkpoint molecules, which plays a role in regulating immune responses. Its expression is elevated in several types

of cancer, including breast cancer, where it has proven to be an effective biomarker for differentiating between normal, benign, precursor, and malignant breast pathologies using UMI [29].

1) *Preparation of B7-H3 Targeted Microbubbles*: B7-H3 targeted microbubbles were prepared using functionalized microbubble formulations purchased in liquid form (USphere Labeler LA, Trust Biosonics, Tapei) and stored at 4°C. The biotinylated antibody targeting B7-H3 was prepared by conjugating monoclonal antibodies (mAbs) with NHS-PEG4-Biotin (ThermoFisher Scientific) at a 1:2 molar ratio. NHS-PEG4-Biotin was dissolved in phosphate-buffered saline (PBS) and incubated with mAbs (1 mg/mL in UltraPure Water) at room temperature for 30 minutes. Unconjugated biotin was removed using a 7 kDa molecular weight cut-off Zeba spin column. For targeted microbubble preparation, 1×10^8 microbubbles/mL/vial were mixed with 10 μ g of the biotinylated B7-H3 mAb, incubated for 1 hour at room temperature, and washed three times with PBS using a microcentrifuge at 300 g for 3 minutes. The topmost floating microbubbles were carefully separated and resuspended in fresh PBS. A final microbubble concentration of 10^7 microbubbles/mL was used after dilution with saline. Non-targeted microbubbles made from the same source as above but without the B7-H3 antibody were used in two mice as a negative control.

2) *Imaging Setup*: B7-H3 targeted microbubbles were injected in ten mice through the mouse tail vein and allowed to circulate for 3 minutes to ensure adequate binding to the B7-H3 target and to clear some of the free microbubbles. A Verasonics Vantage 256 research scanner with an L12-3v transducer was used to collect in vivo RF channel data. We acquired 132 image frames, then burst the bubbles by sending a strong acoustic pulse, and acquired the same number of frames afterward. Each acquisition included 25 plane wave steering angles uniformly distributed over $[-5^\circ, 5^\circ]$ angular range. For each steering angle, a 10 MHz pulse was transmitted to reconstruct the B-mode image, while two pulses at a fundamental frequency of 5 MHz with opposite polarities were transmitted for CEUS imaging. The received signals from the 5 MHz pulses were summed to perform pulse inversion CEUS imaging, targeting the second harmonic at 10 MHz. To generate DTE images, the average of the CEUS frames after the burst is subtracted from a moving average of every 20 frames before the burst, creating corresponding DTE time-series.

3) *Applying the RPCA method*: For the in vivo data, 20, 60, and 132 frames were used to construct the data matrix X and then solve for the low-rank and sparse matrices L and S . The RPCA filter applied to 132 frames was used as a reference, as using more frames improves accuracy of the low rank image at the cost of longer running times. For this reference, to further filter out respiratory motion, we excluded frames with a correlation value below 0.8 relative to their neighboring frames, and a temporal moving average filter with a window of 20 frames was applied to the data.

F. Evaluation Metrics

For the simulation data, the root-mean-square error (RMSE) of the RF data and Dice coefficient of the CEUS images are

used to compare the estimated low-rank and sparse matrices with the ground truth images. The CEUS images were converted to binarized versions after histogram matching, and using a threshold based on the average intensity of the pixels at the bubble locations in the ground-truth image. For the in vivo data, the dice coefficient is used to compare the results with binary versions of the reference and the DTE images. To convert the images to binary, the average intensity of the samples within a 3×3 mm² window in the microbubble region was used as the threshold value. For images of size $m \times n$ and t frames over time, these metrics are defined as follows:

- Root-mean-square error:

$$\text{RMSE} = \sqrt{\frac{1}{mnk} \sum_{t=1}^k \sum_{j=1}^n \sum_{i=1}^m (\hat{c}_{ijt} - c_{ijt})^2}, \quad (7)$$

where c and $\hat{c}$ represent the ground truth and estimated values at image sample (i, j) and frame k over time, calculated separately for the low-rank and sparse matrices.

- Dice Similarity Coefficient:

$$\text{Dice}(C, \hat{C}) = \frac{2|C \cap \hat{C}|}{|C| + |\hat{C}|}, \quad (8)$$

where C and $\hat{C}$ are the estimated and ground-truth (or reference) binary images.

The Dice coefficient is a commonly used metric for evaluating the overlap between two segmented regions, but it is sensitive to class imbalance. In the case of non-targeted microbubbles, the number of microbubbles is significantly lower than the background samples, creating a highly imbalanced dataset where the background dominates. When one class (background) is much larger than the other (microbubbles), the Dice coefficient may not accurately compare DTE and low-rank images. Therefore, the Dice coefficient is not measured for the in vivo cases with non-targeted microbubbles.

III. RESULTS

A. Simulation

An example frame from the time series of one of the simulated samples is shown in Figure 1, where 20 frames were used to construct the data matrix X . The original contrast image in (a) is decomposed into low-rank and sparse matrices, as shown in (b) and (d). The low-rank matrix represents the stationary microbubbles, while the sparse matrix captures the moving microbubbles. Their corresponding ground-truth images are shown in (c) and (e). The Dice coefficients are 0.82 and 0.81 for the low-rank and sparse matrices, respectively, and the corresponding RMS errors are 0.14 and 0.05.

The RMS error and Dice coefficient across ten simulations are calculated and presented in Figure 2 (a)-(b) as a function of the number of temporal frames used in the RPCA filter. The conventional SVD algorithm was also applied to the same data to provide the low-rank matrix as a benchmark, and the residual $X - L$ was considered the sparse matrix. For the low-rank matrix, the RMS error of the SVD method is lower than that of RPCA's low-rank matrix (panel (a)). However, the Dice value for RPCA is higher than that of SVD (panel (b)). For

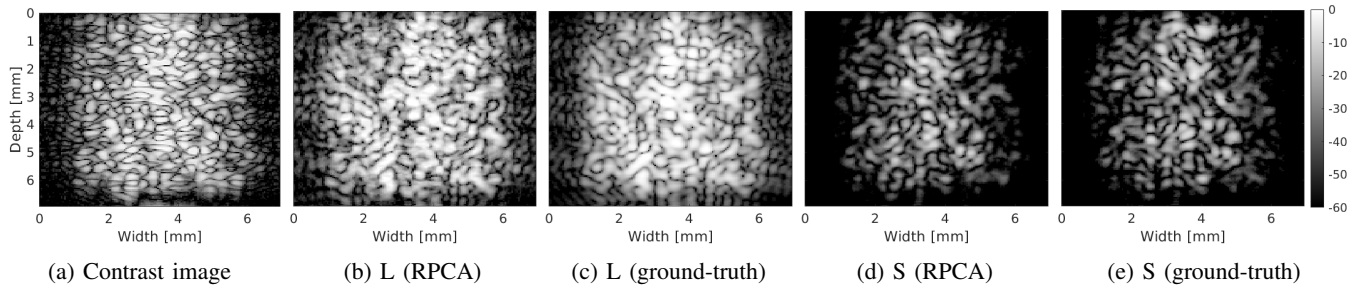


Fig. 1: Representative image from a time-series simulation of stationary and moving microbubbles. The original contrast image in (a) is decomposed into low-rank and sparse matrices, shown in (b) and (d), which represent the stationary and moving microbubbles, respectively. Their corresponding ground-truth images are shown in (c) and (e).

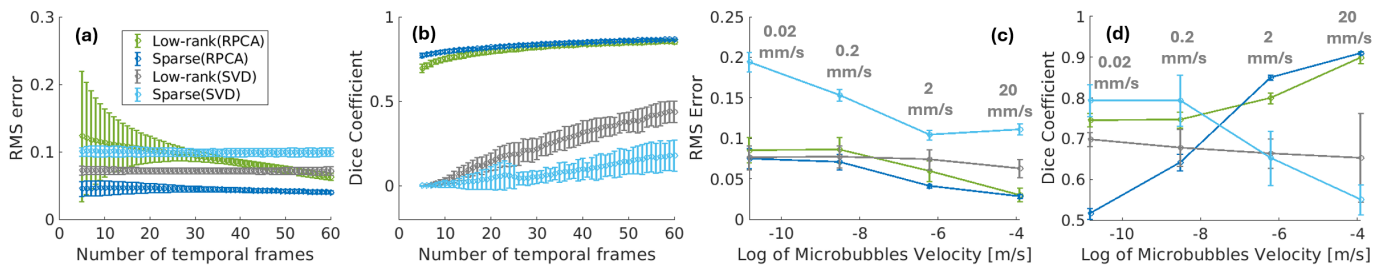


Fig. 2: Simulation results. The RMS error (a) and Dice coefficient (b) as a function of the number of frames over time were calculated between the low-rank and sparse matrices and their corresponding ground-truth images for both the proposed RPCA method and conventional SVD algorithm. The free microbubbles in (a)–(b) have random velocities and directions, with speeds ranging from 0.01 to 2 mm/s. The RMS error (c) and Dice coefficient (d) as a function of the microbubbles velocities were calculated between the low-rank and sparse matrices, obtained from the proposed RPCA and conventional SVD methods, and their corresponding ground-truth images. The horizontal axis represents the logarithm of the microbubble velocity, with the corresponding velocity values in mm/s displayed above the points within the plot. The free microbubbles in (c)–(d) have constant velocity magnitudes with random directions. RPCA and SVD were applied using 20 frames.

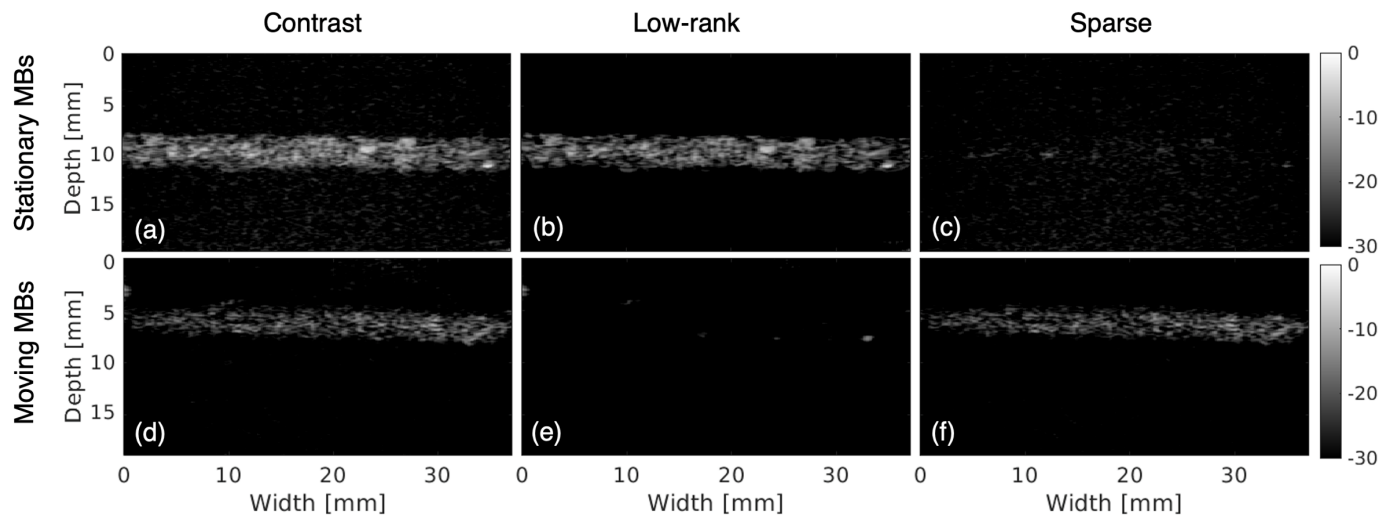


Fig. 3: Phantom experiments showing contrast images of (a-c) stationary and (d-f) moving microbubbles. Top row: (a) Contrast image of non-targeted microbubbles in gelatin injected into a channel. The contrast image was decomposed into a (b) low rank and (c) sparse image. The stationary microbubbles appear in the low rank image with little or no bubble signal in the sparse image, as expected. (d) Contrast image of moving microbubbles. The contrast image was decomposed into a (e) low rank and (f) sparse image. No signal appears in the low rank image and the moving microbubbles only appear in the sparse image.

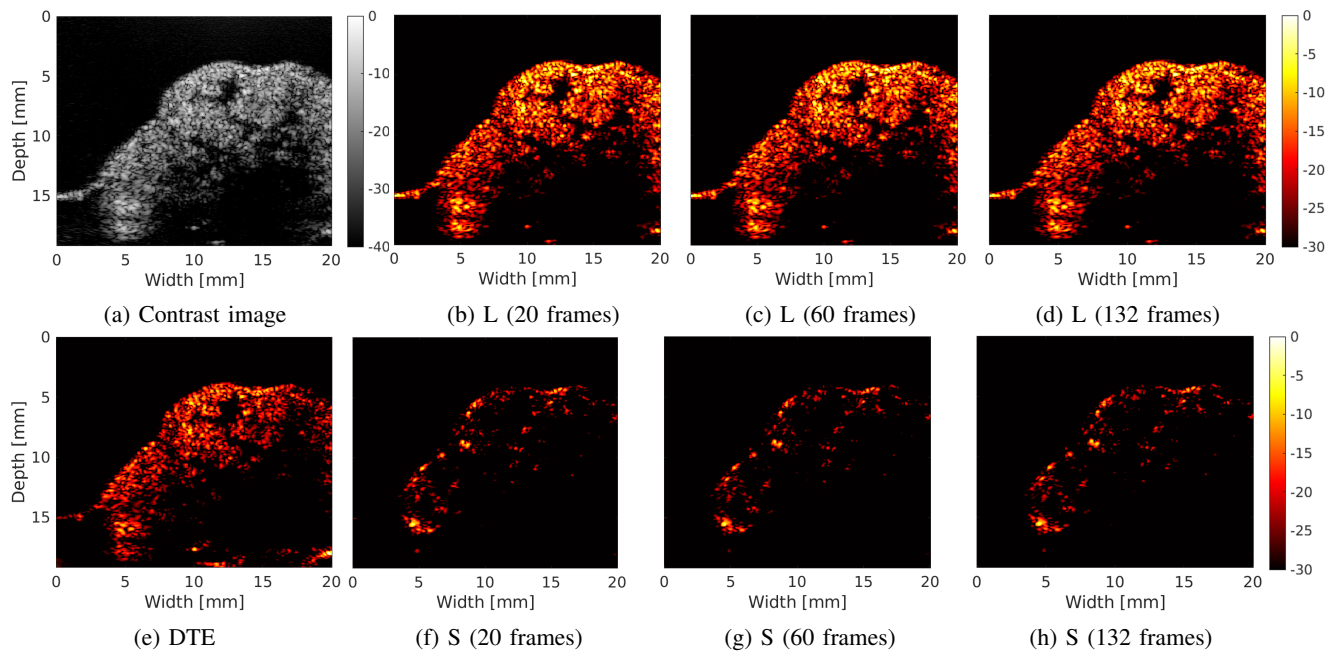


Fig. 4: Representative example from an in vivo experiment with targeted microbubbles. (a) and (e) show the contrast image and the DTE result for bound bubbles. The low-rank images obtained from RPCA using 20, 60, and 132 frames are shown in (b)–(d). The corresponding sparse images from RPCA are shown in (f)–(h).

the sparse component, both the RMS error and Dice values of RPCA are better than those of SVD. The reason is that SVD gives the best low-rank approximation in the least-squares (Frobenius-norm) sense for a chosen rank k (determined here by an energy cutoff); that is, SVD minimizes $\|X - L\|_F$ subject to $\text{rank}(L) = k$. This often leads to a lower RMS error for the low-rank component because SVD allows some sparse energy to leak into the low-rank part to reduce the reconstruction error. However, this leakage degrades the separation, and some sparse targets are partly captured in L , which results in a lower Dice value. RPCA explicitly enforces a decomposition into a low-rank component and a sparse component. This makes RPCA more effective at separating free microbubbles (sparse) from bound microbubbles (low-rank), leading to higher Dice values (better detection or segmentation of the sparse part) and lower RMSE for the sparse component. SVD's low-rank component is obtained by truncation (via an energy cutoff), so it requires more frames to form a stable low-rank subspace. Consequently, the Dice coefficient for SVD improves with frame count but is more variable. In contrast, RPCA's robust objective enables effective component separation even with fewer frames. In the RPCA results, for the low-rank matrix, the RMS error decreases as more frames are used, whereas for the sparse matrix, it remains relatively constant regardless of the number of frames. This is due to thermal noise, which causes the RMSE to reach a lower bound, as low-amplitude thermal noise is never fully reconstructed. Based on the RMS error and Dice coefficient plots for the low-rank matrix, using 20 frames to reconstruct the matrix L with RPCA yields an RMSE of 0.1 and a Dice coefficient of 0.75.

Figure 2 (c)–(d) shows the results of ten simulations, each one with four different velocities for free microbubbles. Panel

(c) presents the RMS error, which decreases as the velocity of the free microbubbles increases for both the proposed RPCA and SVD methods. At lower speeds (below 0.2 mm/s), the low-rank approximations of RPCA and SVD yield similar RMS errors, whereas at higher speeds (2 and 20 mm/s), RPCA achieves lower RMS errors. The sparse component obtained from RPCA is more accurate than that from SVD, since SVD is not optimized to compute the sparse matrix; here, the S matrix for SVD is represented by $X - L$. Panel (d) shows the Dice coefficient. As the microbubble velocity increases, the Dice values increase for the low-rank and sparse components of RPCA and remain approximately constant for the low-rank component of SVD. Both the low-rank and sparse components of RPCA achieve higher Dice values compared to the corresponding components from the SVD method. Since SVD is not optimized to compute a sparse matrix, its sparse component does not perform as well as expected.

B. Phantom

In Figure 3, the top row shows the application of RPCA filtering to stationary microbubbles in a phantom. Panel (a) shows the contrast mode image, where the gel-mixed microbubbles are visible inside the channel. A time series of 20 contrast-mode images is used in the RPCA algorithm to generate time series of low-rank and sparse images, which are shown in panels (b) and (c), respectively. Because most of the bubbles are stationary, they appear in the low-rank image as expected, while the sparse image mostly captures ultrasound noise. The second row shows the application of RPCA filtering to moving microbubbles in a phantom. Panels (d)–(f) present the original contrast mode image, low-rank image, and sparse image, respectively. As expected, the moving microbubbles

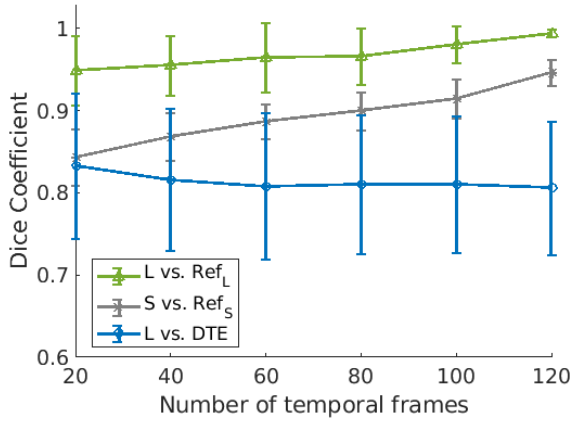


Fig. 5: Dice values for the in vivo experimental results across 10 different mice for the low-rank and sparse matrices versus number of temporal frames compared to the corresponding reference values and DTE.

are present only in the sparse image. Movies showing the time series contrast mode images and corresponding low-rank and sparse images for these two cases are included in the supplementary material.

C. In Vivo

Figure 4 shows a representative example of the in vivo molecular images targeting B7-H3 in breast cancer. Panels (a) and (e) display the original contrast image and DTE images, respectively. The low-rank images obtained from the proposed RPCA method using different numbers of frames are shown in (b)–(d). The corresponding sparse images from RPCA are shown in (f)–(h). Figure 5 shows the average Dice coefficient across 10 mice using the RPCA filter on 132 frames (L: green; S: gray) and DTE (blue) as the reference. For RPCA-filtered images, the average Dice coefficients for the low-rank and sparse components increase from 0.94 to 0.99 and from 0.84 to 0.94, respectively, for increasing number of frames. The average Dice coefficient comparing the low-rank images of RPCA with DTE is relatively constant. The Dice value between the reference low-rank image using 132 frames and the DTE image is 0.80 ± 0.07 (not shown in the plot), indicating good agreement between the two methods.

Figure 6 shows the results of injecting non-targeted microbubbles, with the first and second rows corresponding to two different mice. For each mouse, the original contrast image, the DTE image, the estimated low-rank image, and the sparse image obtained from RPCA are demonstrated. Because non-targeted microbubbles do not bind to molecular targets, they remain freely circulating in the bloodstream and do not accumulate in tumor regions. This results in little to no signal in the tumor area compared to targeted microbubbles as shown in both DTE and low-rank images. Movies showing the time series contrast mode images and corresponding low-rank and sparse components for several in vivo mouse examples are included in the supplementary material.

Figure 7 shows the average signal intensity of the L and

S images within the tumor region across all mice. The L and S images follow the expected characteristics of bound and unbound microbubbles: for the targeted microbubbles, the signal intensity in the L image is higher than in the S image, as the former represents bound microbubbles while the latter represents free microbubbles. For the non-targeted case, the opposite trend is observed, as expected. Furthermore, the signal intensity of the targeted microbubbles is higher in both the L and S images compared to the non-targeted case.

D. Execution Time

There is a trade-off between the number of frames used in the estimation of L and S matrices, the resulting accuracy, and the computational time. While a higher number of frames improves accuracy, it also increases the algorithm's execution time. The average running time to decompose a data matrix consisting of 20 frames, each of size 338 by 260 pixels, into time series of low-rank and sparse matrix sequences of the same size is 0.2 s on a single NVIDIA GeForce RTX 3090 GPU. The average running times for 60 frames and 132 frames are 0.3 s and 0.4 s, respectively.

IV. DISCUSSION

One of the bottlenecks for real-time and nondestructive UMI is separating the free and bound microbubbles, as detecting the latter is the main objective. In this work, we proposed using the RPCA method for this purpose and validated it using simulation, phantom, and in vivo data. Although the method has been demonstrated in the context of molecular imaging of breast cancer, its underlying principles are not limited to this specific application. Since the approach focuses on distinguishing free and bound microbubbles, it can potentially be adapted for molecular imaging of other organs, such as the liver, kidneys, or cardiovascular system.

Molecular readout is based on quantifying signal amplitude. The presence of moving microbubbles can lead to an erroneous elevation in signal amplitude in areas, thereby increasing the risk of false positive cases. In clinical practice, the signal derived from bound microbubbles is of particular interest. Typically, a wait time is required for the moving microbubbles to clear [30], but this wait time varies based on factors such as the type of microbubble, the vascularity of the tissue, and the specific organ being imaged. Furthermore, there is no guarantee that all moving microbubbles will clear out effectively. The proposed RPCA method allows for the nondestructive removal of free microbubbles from the images, resulting in more accurate measurements of the molecular signal. This advancement can enhance early detection of cancer and facilitate monitoring of its progress more reliably.

In the simulation experiment, a random uniform distribution was employed to simulate the initial velocities of the microbubbles, as this method captures a broad spectrum of speeds that may realistically occur in biological environments. This choice accounts for the natural variability observed in microbubble behavior within biological fluids and reflects the stochastic nature of their interactions with surrounding tissues.

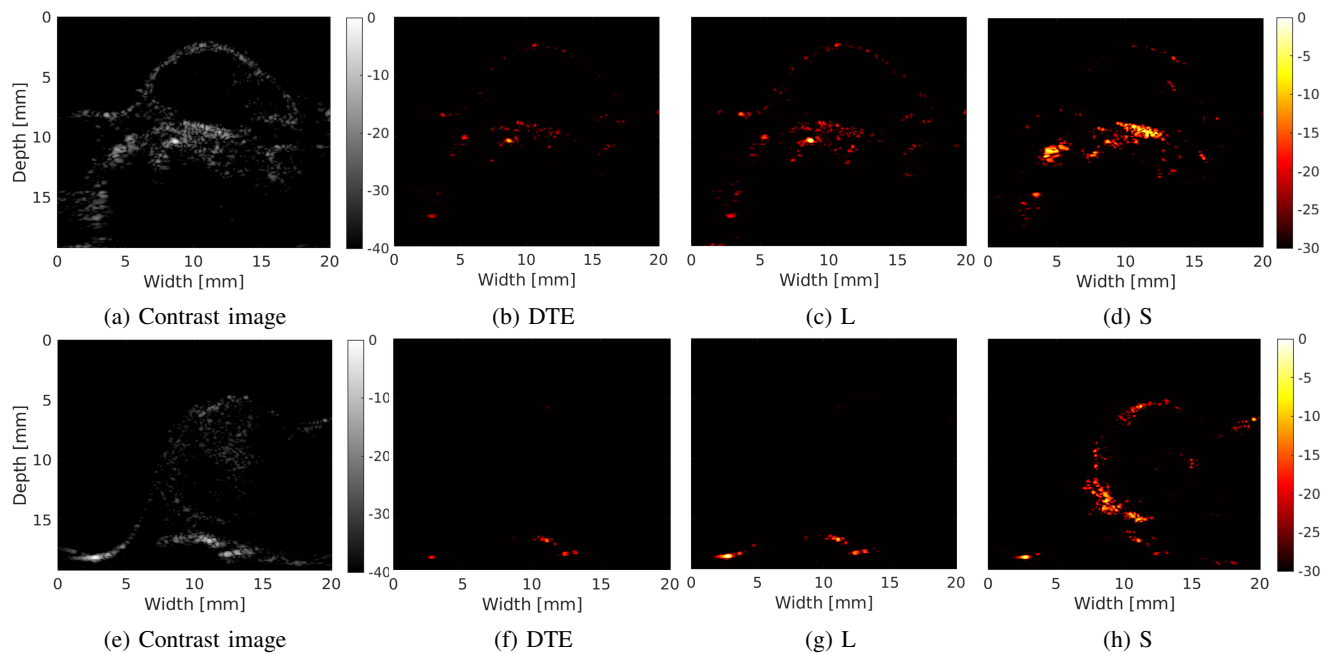


Fig. 6: In vivo experiment with non-targeted bubbles. The first and second rows show data from two different mice, displaying (from left to right) the original harmonic, DTE, RPCA low-rank, and RPCA sparse images.

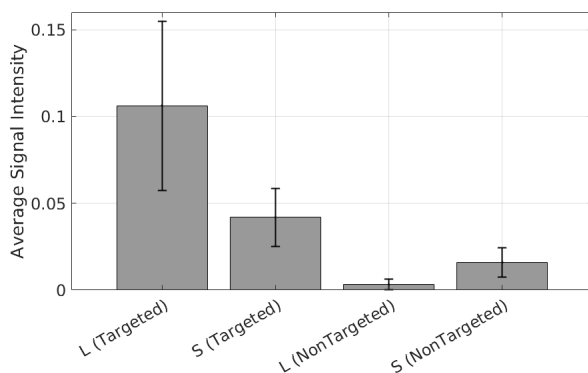


Fig. 7: Average signal intensity of L and S images for targeted and non-targeted microbubbles.

To introduce directionality, sine functions with randomly selected angles were applied, enabling the simulation of curved trajectories and more complex motion patterns. Although this method can result in neighboring microbubbles having notably different velocities and paths, such variability is consistent with what can be observed in vivo. Variations in bubble size, local flow dynamics, and interactions with nearby structures can all significantly influence microbubble movement.

The Dice coefficients for SVD in Figure 2(d), where the microbubbles share the same velocity magnitude, are higher than in Figure 2(b), where the free microbubbles have random velocities and accelerations. This improvement arises because identical velocities yield more predictable trajectories and greater temporal coherence, in contrast to the fully random-velocity case in panels (a)–(b). As a result, the motion energy becomes slightly more concentrated in the singular spectrum, which explains the improvement in SVD performance (higher

Dice values in panel (d)). However, because the bubble directions remain random across bubbles, the motion field is still not globally coherent. In this regime, increasing the velocity magnitude does not make the moving component concentrated in a few singular values; instead, it increases how quickly bubbles cover different locations over time, further distributing their energy across the singular spectrum. Consequently, this constant velocity magnitude scenario remains suboptimal for SVD separation, particularly when only 20 frames are used. As shown in panel (b), SVD performance improves with more frames, which is expected given its strong dependence on ensemble length and the need for many frames to estimate a stable low-rank subspace. With only 20 frames and a sparse component lacking full coherence, the singular spectrum becomes noisy, causing some moving-bubble energy to fall within the top 95%, regardless of the cutoff used. In contrast, the proposed RPCA method decomposes the data into low-rank and sparse components without relying on temporal coherence or energy concentration, enabling it to isolate moving microbubbles reliably and maintain stable performance (Dice > 0.75) even under random-velocity conditions.

In the RPCA method, there is a trade-off between the number of frames used in reconstruction, accuracy, and run time. Using more frames yields more accurate results at the cost of longer running times. For simulation data, the RMS error and Dice coefficient improve as the number of frames used in the estimation of L and S matrices increases. The RMS error of the S matrix is lower than that of the L matrix due to the smaller number of microbubbles and a higher proportion of zero values in the image. The Dice coefficient of S is slightly better than that of L for the same reason. Increasing the number of frames decreases both the average and standard deviation (STD) of the RMS error for the L

matrix. However, for the S matrix, only the STD decreases, while the average RMS error remains relatively constant. This is because ultrasound thermal noise can appear in the S image, limiting the lower bound on the reconstruction error.

For the in vivo data, although using more frames improves accuracy, it also increases susceptibility to tissue or probe motion. If the breathing motion affects the entire background slowly and coherently over time, it is captured in the low-rank component (L). In this work, for the reference results, we eliminated motion, such as respiratory motion, before applying the RPCA filter. As future work, performing motion compensation before RPCA to align frames and reduce the impact of breathing motion, or using high-pass temporal filters before RPCA to remove slow-varying background motion, might be helpful and can be compared with the current results.

The proposed method requires a sufficient number of temporal frames to achieve good performance. Using too few frames (e.g., fewer than 5 frames) results in high RMSE and low Dice values, making detection less reliable. Furthermore, there is a trade-off between accuracy and computation: More frames improve separation accuracy but increase computational cost and memory usage, as the singular value thresholding step in RPCA is computationally expensive, especially with larger frame counts. Furthermore, RMSE is computed on the original-valued data and is therefore invariant to binarization. However, Dice coefficient as a quality metric reflects a trade-off: increasing the threshold reduces false positives but also increases false negatives, while lowering it has the opposite effect. In practice, Dice typically exhibits a broad maximum at an intermediate threshold. To reduce arbitrariness, we selected thresholds from the data (simulations: mean intensity at ground-truth bubble locations; in vivo: mean intensity within a 3×3 mm² microbubble region).

Based on the simulation results, using 20 frames to construct the data matrix is a practical choice, balancing Dice (0.75) and RMSE (0.1) while keeping computational time (0.2 s) and motion manageable. If computational cost is not of concern, increasing the number of frames to 30–40 could slightly improve filtering of the image. The simulation results showed that using a higher number of frames improved separation of the stationary signals from the moving signals.

The proposed method is non-invasive and can remove the moving microbubbles, but it cannot fully eliminate tissue leakage signal, as it appears in the contrast image background and decreases the Dice value (Figure 6 (c)). DTE effectively cancels tissue leakage but it cannot work in real-time and destroys the microbubbles. Furthermore, reducing the number of frames in the averaging for DTE makes it more susceptible to moving bubbles.

A limitation of the in vivo study is the absence of a ground-truth for the experiments. To address this, we used a large number of frames (132) and removed those affected by motion before applying RPCA, which was then used as a reference because it achieves better separation of stationary and free microbubbles, as demonstrated in the simulation results (Figure 2). This reference does not completely eliminate tissue leakage signals and its dataset covers different temporal frames compared to DTE. These differences explain the lower

Dice value relative to DTE compared with that of the 20-frame RPCA. Despite this, it shows good agreement with DTE, and the differences between using 132 frames versus 20 frames were relatively small. DTE and 20-frame RPCA show a good match, as both methods cover the same temporal window (20 frames), but the Dice value is affected by the tissue leakage present in the 20-frame RPCA image and the incomplete removal of free microbubbles in the DTE image. In summary, the 20-frame RPCA was designed for semi-real-time implementation, being less computationally demanding and less susceptible to motion, whereas the 132-frame RPCA was used to achieve the best separation of stationary and free microbubbles regardless of processing time. A potential solution to these limitations would be to integrate the RPCA filter into our prior neural network-based approach [9] to leverage the benefits of tissue leakage cancellation while also eliminating moving microbubbles.

The GPU-based implementation of RPCA enabled efficient decomposition of UMI data, significantly reducing computation time compared to a CPU-based approach. However, the RPCA algorithm's computational complexity grows with both the number of frames and the spatial resolution of the imaging matrix. In our current implementation, using a matrix of size 338 by 260 pixels and 20 frames, the GPU-based decomposition completes in approximately 0.2 s on an NVIDIA RTX 3090. Larger matrices would proportionally increase computation time, primarily due to the singular value decomposition step. However, modern clinical scanners equipped with powerful GPUs or dedicated hardware accelerators (e.g., tensor cores) are increasingly capable of handling such loads. Furthermore, applying block-wise or region-of-interest-based RPCA could help manage computational demand in larger matrices. Although the execution time of 0.2 s/frame series suggests that real-time performance (on the order of 5 frames/s or faster) is feasible, further optimization could improve performance and enable higher frame rates. While our implementation leverages CUDA acceleration through PyTorch, it primarily focuses on functionality rather than full optimization. Techniques such as mixed-precision computation, kernel fusion, and memory-efficient tensor operations could further accelerate processing [31], [32] to achieve real-time performance. Additionally, implementing a more hardware-aware approach, such as optimizing memory access patterns and utilizing tensor cores, could enhance performance [31], [33]. These enhancements are common in GPU programming and could be integrated into future iterations of the software to support real-time clinical imaging.

In our in vivo studies, although the RPCA method demonstrated consistent performance across ten transgenic mice, translating these findings to clinical practice may present challenges, including increased physiological motion and variability in signal intensity due to differences in vascular structures, microbubble targets, or organs. Furthermore, real-time implementation will require system integration efforts to synchronize the RPCA algorithm with the imaging script.

V. CONCLUSION

We proposed a fast GPU-based method leveraging robust principal component analysis to differentiate bound microbubbles from free-floating ones. The method was validated using ten simulation datasets and phantom experiments. Furthermore, it was applied to data from ten transgenic mouse models of breast cancer development injected with B7-H3-targeted microbubbles, and two mice injected with non-targeted microbubbles. The results align well with the reference results, with a Dice coefficient higher than 0.95, and with those of the conventional differential targeted enhancement method, which shows a Dice value higher than 0.8. The average running time of the method on a data matrix of size $338 \times 260 \times 20$ was 0.2 s on a single GPU.

REFERENCES

- [1] A. L. Klibanov, "Ligand-carrying gas-filled microbubbles: ultrasound contrast agents for targeted molecular imaging," *Bioconjugate chemistry*, vol. 16, no. 1, pp. 9–17, 2005.
- [2] N. Deshpande, A. Needles, and J. K. Willmann, "Molecular ultrasound imaging: current status and future directions," *Clinical radiology*, vol. 65, no. 7, pp. 567–581, 2010.
- [3] R. J. Eckersley, C. T. Chin, and P. N. Burns, "Optimising phase and amplitude modulation schemes for imaging microbubble contrast agents at low acoustic power," *Ultrasound in medicine & biology*, vol. 31, no. 2, pp. 213–219, 2005.
- [4] P. Phillips, "Contrast pulse sequences (CPS): Imaging nonlinear microbubbles," in *2001 IEEE Ultrasonics Symposium. Proceedings. An International Symposium (Cat. No. 01CH37263)*, vol. 2. IEEE, 2001, pp. 1739–1745.
- [5] G. Collado-Lara, G. Wahyulaksana, H. J. Vos, and K. Kooiman, "Non-destructive ultrasound molecular imaging with high-order singular value decomposition," *Authorea Preprints*, 2024.
- [6] J. R. Lindner, J. Song, J. Christiansen, A. L. Klibanov, F. Xu, and K. Ley, "Ultrasound assessment of inflammation and renal tissue injury with microbubbles targeted to p-selectin," *Circulation*, vol. 104, no. 17, pp. 2107–2112, 2001.
- [7] P. A. Dayton and J. J. Rychak, "Molecular ultrasound imaging using microbubble contrast agents," *Front Biosci*, vol. 12, no. 23, pp. 5124–5142, 2007.
- [8] L. Abou-Elkacem, S. V. Bachawal, and J. K. Willmann, "Ultrasound molecular imaging: Moving toward clinical translation," *European journal of radiology*, vol. 84, no. 9, pp. 1685–1693, 2015.
- [9] D. Hyun, L. Abou-Elkacem, R. Bam, L. L. Brickson, C. D. Herickhoff, and J. J. Dahl, "Nondestructive detection of targeted microbubbles using dual-mode data and deep learning for real-time ultrasound molecular imaging," *IEEE transactions on medical imaging*, vol. 39, no. 10, pp. 3079–3088, 2020.
- [10] M. A. Pysz, I. Guracar, L. Tian, and J. K. Willmann, "Fast microbubble dwell-time based ultrasonic molecular imaging approach for quantification and monitoring of angiogenesis in cancer," *Quantitative imaging in medicine and surgery*, vol. 2, no. 2, p. 68, 2012.
- [11] V. Daeichin, Z. Akkus, I. Skachkov, K. Kooiman, A. Needles, J. Sluimer, B. Janssen, M. J. Daemen, A. F. Van Der Steen, N. De Jong *et al.*, "Quantification of bound microbubbles in ultrasound molecular imaging," *IEEE transactions on ultrasonics, ferroelectrics, and frequency control*, vol. 62, no. 6, pp. 1190–1200, 2015.
- [12] F. W. Mauldin, A. H. Dhanaliwala, A. V. Patil, and J. A. Hossack, "Real-time targeted molecular imaging using singular value spectra properties to isolate the adherent microbubble signal," *Physics in Medicine & Biology*, vol. 57, no. 16, p. 5275, 2012.
- [13] E. B. Herbst, S. Unnikrishnan, A. L. Klibanov, F. W. Mauldin Jr, and J. A. Hossack, "Validation of normalized singular spectrum area as a classifier for molecularly targeted microbubble adherence," *Ultrasound in medicine & biology*, vol. 45, no. 9, pp. 2493–2501, 2019.
- [14] K. Xu, X. Guo, Y. Sui, V. Hingot, O. Couture, D. Ta, and W. Wang, "Robust PCA-based clutter filtering method for super-resolution ultrasound localization microscopy," in *2021 IEEE International Ultrasonics Symposium (IUS)*. IEEE, 2021, pp. 1–4.
- [15] H. S. Hashemi, D. Hyun, J. Baek, A. Natarajan, F. Tabesh, R. Paulmurugan, and J. J. Dahl, "Enhancing ultrasound molecular imaging: RPCA-based filtering to differentiate tumor-bound and free microbubbles," in *2024 IEEE Ultrasonics, Ferroelectrics, and Frequency Control Joint Symposium (UFFC-JS)*. IEEE, 2024, pp. 1–6.
- [16] E. J. Candès, X. Li, Y. Ma, and J. Wright, "Robust principal component analysis?" *Journal of the ACM (JACM)*, vol. 58, no. 3, pp. 1–37, 2011.
- [17] X. Yuan and J. Yang, "Sparse and low-rank matrix decomposition via alternating direction methods," *Optimization Online*, vol. 12, no. 2, 2009.
- [18] J. Reinhold, "Implementations of RPCA on the GPU," <https://gist.github.com/jcreinhold>, 2019.
- [19] J. A. Jensen, "Field: A program for simulating ultrasound systems," *Medical & Biological Engineering & Computing*, vol. 34, no. sup. 1, pp. 351–353, 1997.
- [20] J. A. Jensen and N. B. Svendsen, "Calculation of pressure fields from arbitrarily shaped, apodized, and excited ultrasound transducers," *IEEE transactions on ultrasonics, ferroelectrics, and frequency control*, vol. 39, no. 2, pp. 262–267, 1992.
- [21] P. Marmottant, S. Van Der Meer, M. Emmer, M. Versluis, N. De Jong, S. Hilgenfeldt, and D. Lohse, "A model for large amplitude oscillations of coated bubbles accounting for buckling and rupture," *The Journal of the Acoustical Society of America*, vol. 118, no. 6, pp. 3499–3505, 2005.
- [22] L. Hoff, *Acoustic characterization of contrast agents for medical ultrasound imaging*. Springer Science & Business Media, 2001.
- [23] R. H. Abou-Saleh, F. J. Armistead, D. V. Batchelor, B. R. Johnson, S. A. Peyman, and S. D. Evans, "Horizon: Microfluidic platform for the production of therapeutic microbubbles and nanobubbles," *Review of Scientific Instruments*, vol. 92, no. 7, 2021.
- [24] S. S. Brunke, M. F. Insana, J. J. Dahl, C. Hansen, M. Ashfaq, and H. Ermert, "An ultrasound research interface for a clinical system," *IEEE transactions on ultrasonics, ferroelectrics, and frequency control*, vol. 54, no. 1, pp. 198–210, 2007.
- [25] H. Ishida, T. Andoh, S. Akiguchi, T. Kyoden, and T. Hachiga, "Observation of tumor microvessels that are controlled by blood flow in breast cancer," *Applied Physics Letters*, vol. 106, no. 17, 2015.
- [26] J. Li, C. Wei, T. Ying, Y. Liu, R. Wang, M. Li, C. Feng, D. Sun, and Y. Zheng, "Differentiation of benign and malignant breast lesions by ultrasound localization microscopy," *Insights into Imaging*, vol. 16, no. 1, p. 128, 2025.
- [27] W.-J. Lee, J.-S. Chu, C.-S. Huang, M.-F. Chang, K.-J. Chang, and K.-M. Chen, "Breast cancer vascularity: color doppler sonography and histopathology study," *Breast cancer research and treatment*, vol. 37, pp. 291–298, 1996.
- [28] E. Y. Lin, J. G. Jones, P. Li, L. Zhu, K. D. Whitney, W. J. Muller, and J. W. Pollard, "Progression to malignancy in the polyoma middle t oncoprotein mouse breast cancer model provides a reliable model for human diseases," *The American journal of pathology*, vol. 163, no. 5, pp. 2113–2126, 2003.
- [29] S. V. Bachawal, K. C. Jensen, K. E. Wilson, L. Tian, A. M. Lutz, and J. K. Willmann, "Breast cancer detection by B7-H3-targeted ultrasound molecular imaging," *Cancer research*, vol. 75, no. 12, pp. 2501–2509, 2015.
- [30] J. K. Willmann, L. Bonomo, A. C. Testa, P. Rinaldi, G. Rindi, K. S. Valluru, G. Petrone, M. Martini, A. M. Lutz, and S. S. Gambhir, "Ultrasound molecular imaging with br55 in patients with breast and ovarian lesions: first-in-human results," *Journal of Clinical Oncology*, vol. 35, no. 19, pp. 2133–2140, 2017.
- [31] P. Mickevicius, S. Narang, J. Alben, G. Diamos, E. Elsen, D. Garcia, B. Ginsburg, M. Houston, O. Kuchaiev, G. Venkatesh *et al.*, "Mixed precision training," *arXiv preprint arXiv:1710.03740*, 2017.
- [32] T. Chen, T. Moreau, Z. Jiang, L. Zheng, E. Yan, H. Shen, M. Cowan, L. Wang, Y. Hu, L. Ceze *et al.*, "TVM: An automated end-to-end optimizing compiler for deep learning," in *13th USENIX Symposium on Operating Systems Design and Implementation (OSDI 18)*, 2018, pp. 578–594.
- [33] D. Guide, "Cuda C++ best practices guide," 2020.